Oracle GoldenGate Configuration

```
##############################################################################
## Copyright(c) Oracle Corporation 2019. All rights reserved.                ##
##                                                                           ##
## Specify values for the variables listed below to customize                ##
## your installation.                                                        ##
##                                                                           ##
## Each variable is associated with a comment. The comment                   ##
## can help to populate the variables with the appropriate                   ##
## values.                                                                   ##
##                                                                           ##
## IMPORTANT NOTE: This file should be secured to have read                  ##
## permission only by the oracle user or an administrator who                ##
## own this installation to protect any sensitive input values.              ##
##                                                                           ##
##############################################################################

#------------------------------------------------------------------------------
# Do not change the following system generated value.
#------------------------------------------------------------------------------
oracle.install.responseFileVersion=/oracle/install/rspfmt_ogginstall_response_schema_v19_1_0


###############################################################################################
##                                                                                           ##
## Oracle GoldenGate installation option and details                                         ##
##                                                                                           ##
###############################################################################################

#------------------------------------------------------------------------------
# Specify the installation option.
# Specify ORA19c for installing Oracle GoldenGate for Oracle Database 19c or
#         ORA18c for installing Oracle GoldenGate for Oracle Database 18c or
#         ORA12c for installing Oracle GoldenGate for Oracle Database 12c or
#         ORA11g for installing Oracle GoldenGate for Oracle Database 11g
#------------------------------------------------------------------------------
INSTALL_OPTION=ORA19c
```

```
#-------------------------------------------------------------------------------
# Specify a location to install Oracle GoldenGate
#-------------------------------------------------------------------------------
SOFTWARE_LOCATION=/u02/app/oracle/gghome/19c

#-------------------------------------------------------------------------------
# Specify true to start the manager after installation.
#-------------------------------------------------------------------------------
START_MANAGER=true

#-------------------------------------------------------------------------------
# Specify a free port within the valid range for the manager process.
# Required only if START_MANAGER is true.
#-------------------------------------------------------------------------------
MANAGER_PORT=8706

#-------------------------------------------------------------------------------
# Specify the location of the Oracle Database.
# Required only if START_MANAGER is true.
#-------------------------------------------------------------------------------
DATABASE_LOCATION=/u02/app/oracle/product/19c/db_1


###############################################################################
##                                                                           ##
## Specify details to Create inventory for Oracle installs                   ##
## Required only for the first Oracle product install on a system.           ##
##                                                                           ##
###############################################################################

#-------------------------------------------------------------------------------
# Specify the location which holds the install inventory files.
# This is an optional parameter if installing on
# Windows based Operating System.
#-------------------------------------------------------------------------------
INVENTORY_LOCATION=/u02/app/oracle/oraInventory

#-------------------------------------------------------------------------------
# Unix group to be set for the inventory directory.
# This parameter is not applicable if installing on
# Windows based Operating System.
#-------------------------------------------------------------------------------
UNIX_GROUP_NAME=oinstall
[oracle@ipfix01 response]$
```

```
SQL> ARCHIVE LOG LIST;
Database log mode              No Archive Mode
Automatic archival             Disabled
Archive destination            /u02/app/oracle/product/19c/db_1/dbs/arch
Oldest online log sequence     22
Current log sequence           24
SQL>
```

```
SQL> SHUTDOWN IMMEDIATE;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>
```

STARTUP MOUNT;

```
SQL> STARTUP MOUNT;
ORACLE instance started.

Total System Global Area 2432693152 bytes
Fixed Size                  8942496 bytes
Variable Size             754974720 bytes
Database Buffers         1660944384 bytes
Redo Buffers                7831552 bytes
Database mounted.
SQL>
```

ALTER DATABASE ARCHIVELOG;

```
SQL> ALTER DATABASE ARCHIVELOG;

Database altered.
```

ALTER DATABASE FORCE LOGGING;

ALTER SYSTEM SET log_archive_dest_1='LOCATION=/u01/oracle/archivelogs' SCOPE=SPFILE;

ALTER SYSTEM SET log_archive_format='arch_%t_%s_%r.arc' SCOPE=SPFILE;

```
SQL> ALTER SYSTEM SET log_archive_dest_1='LOCATION=/u02/app/oracle/arch' SCOPE=SPFILE;

System altered.

SQL> ALTER SYSTEM SET log_archive_format='arch_%t_%s_%r.arc' SCOPE=SPFILE;

System altered.

SQL>
```

```
SQL> ALTER DATABASE OPEN;

Database altered.
```

```
SQL> ARCHIVE LOG LIST;
Database log mode              Archive Mode
Automatic archival             Enabled
Archive destination            /u02/app/oracle/arch
Oldest online log sequence     22
Next log sequence to archive   24
Current log sequence           24
```

- # Preparing the Database for Oracle GoldenGate

Configuring Connections for Integrated Processes

If you will be using integrated capture and integrated Replicat, each requires a dedicated server connection in the `tnsnames.ora` file.

```
GGSR =
  (DESCRIPTION =
   (ADDRESS_LIST =
    (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.68.121)(PORT = 1521)))
    (CONNECT_DATA =
     (SERVER = DEDICATED)
      (SERVICE_NAME = GGSR)
     )
  )
[oracle@ipfix01 admin]$
```

```
[oracle@ipfix01 admin]$ tnsping GGSR

TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 25-AUG-2025 11:52:05

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Used parameter files:


Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS_LIST = (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.68.121)(PORT = 1521))) (CONNECT_DATA = (SERVER = DEDICATED) (SERVICE_NAM
E = GGSR)))
OK (10 msec)
[oracle@ipfix01 admin]$
```

```
SQL> CREATE USER ggadmin IDENTIFIED BY password DEFAULT
TABLESPACE GG_DATA TEMPORARY TABLESPACE temp;  2

User created.

SQL> GRANT CREATE SESSION to ggadmin;
GRANT CREATE VIEW to ggadmin;
GRANT CONNECT to ggadmin ;
GRANT RESOURCE to ggadmin;
GRANT ALTER SYSTEM to ggadmin ;
GRANT SELECT ANY DICTIONARY to ggadmin ;
Grant succeeded.

SQL>
Grant succeeded.
```

```
SQL> EXEC DBMS_GOLDENGATE_AUTH.GRANT_ADMIN_PRIVILEGE('ggadmin');

PL/SQL procedure successfully completed.

SQL> ALTER USER ggadmin QUOTA unlimited ON GG_DATA;

User altered.

SQL> SELECT * FROM DBA_SYS_PRIVS WHERE GRANTEE='GGADMIN' ORDER BY 2;

GRANTEE PRIVILEGE                                ADM COM INH
------- ---------------------------------------- --- --- ---
GGADMIN ALTER ANY TABLE                          NO  NO  NO
GGADMIN ALTER SESSION                            NO  NO  NO
GGADMIN ALTER SYSTEM                             NO  NO  NO
GGADMIN CREATE ANY EDITION                       NO  NO  NO
GGADMIN CREATE EVALUATION CONTEXT                YES NO  NO
GGADMIN CREATE JOB                               NO  NO  NO
GGADMIN CREATE RULE                              YES NO  NO
GGADMIN CREATE RULE SET                          YES NO  NO
GGADMIN CREATE SESSION                           NO  NO  NO
GGADMIN CREATE VIEW                              NO  NO  NO
GGADMIN DEQUEUE ANY QUEUE                        YES NO  NO

GRANTEE PRIVILEGE                                ADM COM INH
------- ---------------------------------------- --- --- ---
GGADMIN DROP ANY EDITION                         NO  NO  NO
GGADMIN EXECUTE ANY RULE SET                     YES NO  NO
GGADMIN FLASHBACK ANY TABLE                      NO  NO  NO
GGADMIN LOGMINING                                NO  NO  NO
GGADMIN SELECT ANY DICTIONARY                    NO  NO  NO
GGADMIN SELECT ANY TABLE                         NO  NO  NO

17 rows selected.

SQL>
```

Enabling Minimum Database-level Supplemental Logging

Oracle strongly recommends putting the Oracle source database into forced logging mode. Forced logging mode forces the logging of all transactions and loads, overriding any user or storage settings to the contrary. This ensures that no source data in the Extract configuration gets missed.

Issue the following command to determine whether the database is in supplemental logging mode and in forced logging mode. If the result is `YES` for both queries, the database meets the Oracle GoldenGate requirement.

```
SQL> SELECT supplemental_log_data_min, force_logging FROM v$database;

SUPPLEME FORCE_LOGGING
-------- ---------------------------------------
NO       NO
```

```
ALTER DATABASE ADD SUPPLEMENTAL LOG DATA;

ALTER DATABASE FORCE LOGGING;
```

```
SQL> ALTER DATABASE ADD SUPPLEMENTAL LOG DATA;

Database altered.

SQL> ALTER DATABASE FORCE LOGGING;

Database altered.

SQL> SELECT supplemental_log_data_min, force_logging FROM v$database;

SUPPLEME FORCE_LOGGING
-------- ---------------------------------------
YES      YES

SQL>
```

```
SQL> ALTER SYSTEM SWITCH LOGFILE;

System altered.
```

Enabling Oracle GoldenGate in the Database

The database services required to support Oracle GoldenGate capture and apply must be enabled explicitly for all Oracle database versions.

```
SQL> alter system set ENABLE_GOLDENGATE_REPLICATION=true scope=both;

System altered.
```

```
SQL> show parameter  ENABLE_GOLDENGATE_REPLICATION

NAME                                 TYPE        VALUE
------------------------------------ ----------- -------
enable_goldengate_replication        boolean     TRUE
SQL>
```

TABLE_NAME

DRIVERS
SHIPMENTS
VEHICLES

Enabling Table-level Supplemental Logging

ADD CREDENTIALSTORE
ALTER CREDENTIALSTORE ADD USER ggadmin ALIAS gguser
DBLOGIN USERIDALIAS gguser

```
GGSCI (ipfix01.example.com) 2> ADD CREDENTIALSTORE

Credential store created.

GGSCI (ipfix01.example.com) 3> ALTER CREDENTIALSTORE ADD USER ggadmin ALIAS gguser
Password:

Credential store altered.

GGSCI (ipfix01.example.com) 4>
```

```
GGSCI (ipfix01.example.com) 4> DBLOGIN USERIDALIAS gguser
Successfully logged into database.

GGSCI (ipfix01.example.com as ggadmin@DDSTST) 5>
```

Perform the following steps on the source system to enable table-level supplemental logging or use the optional features of the command.

```
ADD TRANDATA  LOGIUSER.DRIVERS
```

```
GGSCI (ipfix01.example.com as ggadmin@DDSTST) 5> ADD TRANDATA  LOGIUSER.DRIVERS

2025-08-25 12:59:49  WARNING OGG-01896  Table LOGIUSER.DRIVERS has an identity column which is not supported. This table will be ignored by Extract.

2025-08-25 12:59:51  INFO    OGG-15132  Logging of supplemental redo data enabled for table LOGIUSER.DRIVERS.

2025-08-25 12:59:51  INFO    OGG-15133  TRANDATA for scheduling columns has been added on table LOGIUSER.DRIVERS.

2025-08-25 12:59:51  INFO    OGG-15135  TRANDATA for instantiation CSN has been added on table LOGIUSER.DRIVERS.

2025-08-25 12:59:51  WARNING OGG-01896  Table LOGIUSER.DRIVERS has an identity column which is not supported. This table will be ignored by Extract.

2025-08-25 12:59:53  INFO    OGG-10471  ***** Oracle Goldengate support information on table LOGIUSER.DRIVERS *****
Oracle Goldengate support native capture on table LOGIUSER.DRIVERS.
Oracle Goldengate marked following column as key columns on table LOGIUSER.DRIVERS: DRIVER_ID.

GGSCI (ipfix01.example.com as ggadmin@DDSTST) 6>
```

```
GGSCI (ipfix01.example.com) 1> DBLOGIN USERIDALIAS gguser
Successfully logged into database.

GGSCI (ipfix01.example.com as ggadmin@DDSTST) 2>
```

```
DBLOGIN USERIDALIAS gguser
ADD TRANDATA  LOGIUSER.SHIPMENTS
GGSCI (ipfix01.example.com as ggadmin@DDSTST) 2> ADD TRANDATA  LOGIUSER.SHIPMENTS

2025-08-25 13:08:26  WARNING OGG-01896  Table LOGIUSER.SHIPMENTS has an identity
column which is not supported. This table will be ignored by Extract.

2025-08-25 13:08:26  INFO    OGG-15132  Logging of supplemental redo data enabled
for table LOGIUSER.SHIPMENTS.

2025-08-25 13:08:26  INFO    OGG-15133  TRANDATA for scheduling columns has been
added on table LOGIUSER.SHIPMENTS.

2025-08-25 13:08:26  INFO    OGG-15135  TRANDATA for instantiation CSN has been
added on table LOGIUSER.SHIPMENTS.
```

```
2025-08-25 13:08:26  WARNING OGG-01896  Table LOGIUSER.SHIPMENTS has an identity
column which is not supported. This table will be ignored by Extract.

2025-08-25 13:08:27  INFO    OGG-10471  ***** Oracle Goldengate support information
on table LOGIUSER.SHIPMENTS *****
Oracle Goldengate support native capture on table LOGIUSER.SHIPMENTS.
Oracle Goldengate marked following column as key columns on table
LOGIUSER.SHIPMENTS: SHIPMENT_ID.

GGSCI (ipfix01.example.com as ggadmin@DDSTST) 3>

ADD TRANDATA  LOGIUSER.VEHICLES
GGSCI (ipfix01.example.com as ggadmin@DDSTST) 3> ADD TRANDATA  LOGIUSER.VEHICLES

2025-08-25 13:09:18  WARNING OGG-01896  Table LOGIUSER.VEHICLES has an identity
column which is not supported. This table will be ignored by Extract.

2025-08-25 13:09:18  INFO    OGG-15132  Logging of supplemental redo data enabled
for table LOGIUSER.VEHICLES.

2025-08-25 13:09:18  INFO    OGG-15133  TRANDATA for scheduling columns has been
added on table LOGIUSER.VEHICLES.

2025-08-25 13:09:18  INFO    OGG-15135  TRANDATA for instantiation CSN has been
added on table LOGIUSER.VEHICLES.

2025-08-25 13:09:18  WARNING OGG-01896  Table LOGIUSER.VEHICLES has an identity
column which is not supported. This table will be ignored by Extract.

2025-08-25 13:09:18  INFO    OGG-10471  ***** Oracle Goldengate support information
on table LOGIUSER.VEHICLES *****
Oracle Goldengate support native capture on table LOGIUSER.VEHICLES.
Oracle Goldengate marked following column as key columns on table
LOGIUSER.VEHICLES: VEHICLE_ID.
```

Integrated Capture:

In integrated capture mode, the Oracle GoldenGate Extract process interacts directly with a database logmining server to receive data changes in the form of logical change records (LCR).

Configuring the Primary Extract in Integrated Capture Mode

The mining database from which the primary Extract captures log change records from the logmining server, can be either local or downstream from the source database.

Edit param shipep

```
EXTRACT shipep
USERIDALIAS gguser
DDL INCLUDE MAPPED
EXTTRAIL /u02/app/oracle/gghome/19c/dirdat/lt
SEQUENCE LOGIUSER.SHIPMENTS_seq;
TABLE LOGIUSER.*;
```

Configuring the Data Pump Extract

A data pump can perform data filtering, mapping, and conversion, or it can be configured in pass-through mode, where data is passively transferred as-is, without manipulation.
Edit param shippump

```
EXTRACT shippump
USERIDALIAS gguser
RMTHOST ipfix01.example.com, MGRPORT 7809
RMTTRAIL /u02/app/oracle/gghome2/19c/dirdat/rt
SEQUENCE LOGIUSER.SHIPMENTS_seq;
TABLE LOGIUSER.*;
```

Prerequisites for Configuring Replicat

The guidelines to follow before configuring Replicat are:

1. Preparing the Database for Oracle GoldenGate.
2. Establishing Oracle GoldenGate Credentials.
3. Choosing Capture and Apply Modes.
4. Create the Oracle GoldenGate instance on the target system by configuring the Manager process.

```
GGSCI (ipfix01.example.com) 1> ADD CREDENTIALSTORE

Credential store created.

GGSCI (ipfix01.example.com) 2> ALTER CREDENTIALSTORE ADD USER ggadmin2 ALIAS gguser2
Password:

Credential store altered.

GGSCI (ipfix01.example.com) 3> DBLOGIN USERIDALIAS gguser2
Successfully logged into database.

GGSCI (ipfix01.example.com as ggadmin2@DDSTST) 4>
```

Creating a Checkpoint Table

Adding the Checkpoint Table to the Target Database

```
DBLOGIN USERIDALIAS gguser2
ADD CHECKPOINTTABLE GGADMIN2.DRIVERS
ADD CHECKPOINTTABLE GGADMIN2.SHIPMENTS
ADD CHECKPOINTTABLE GGADMIN2.VEHICLES
```

```
GGSCI (ipfix01.example.com) 1> DBLOGIN USERIDALIAS gguser2
Successfully logged into database.

GGSCI (ipfix01.example.com as ggadmin2@DDSTST) 2> ADD CHECKPOINTTABLE LOGIUSER2.DRIVERS
ERROR: Failed creating checkpoint table LOGIUSER2.DRIVERS.

OCI Error ORA (status = 1031-ORA-01031: insufficient privileges
).

GGSCI (ipfix01.example.com as ggadmin2@DDSTST) 3> ADD CHECKPOINTTABLE GGADMIN2.DRIVERS

Successfully created checkpoint table GGADMIN2.DRIVERS.

GGSCI (ipfix01.example.com as ggadmin2@DDSTST) 4> ADD CHECKPOINTTABLE GGADMIN2.SHIPMENTS

Successfully created checkpoint table GGADMIN2.SHIPMENTS.

GGSCI (ipfix01.example.com as ggadmin2@DDSTST) 5> ADD CHECKPOINTTABLE GGADMIN2.VEHICLES

Successfully created checkpoint table GGADMIN2.VEHICLES.

GGSCI (ipfix01.example.com as ggadmin2@DDSTST) 6>
```

Configuring Replicat

```
EDIT PARAM SHIPR
REPLICAT shipr
USERIDALIAS gguser2
ASSUMETARGETDEFS
MAP LOGIUSER.*, TARGET LOGIUSER2.*;
```

Creating Process Groups

Registering Extract with the Mining Database

If you are using Extract in integrated mode, you need to create a database logmining server to capture redo data. You do this from the GGSCI interface by registering the primary Extract process with the mining database.

DBLOGIN USERIDALIAS gguser
REGISTER EXTRACT shipep DATABASE

```
GGSCI (ipfix01.example.com) 3> DBLOGIN USERIDALIAS gguser
Successfully logged into database.

GGSCI (ipfix01.example.com as ggadmin@DDSTST) 4> REGISTER EXTRACT shipep DATABASE

2025-08-25 15:50:23  INFO    OGG-02003  Extract SHIPEP successfully registered with database at SCN 3593212.


GGSCI (ipfix01.example.com as ggadmin@DDSTST) 5>
```

ADD EXTRACT shipep, INTEGRATED TRANLOG, BEGIN NOW
ADD EXTTRAIL /u02/app/oracle/gghome/19c/dirdat/lt, EXTRACT shipep
ADD EXTRACT shippump, EXTTRAILSOURCE /u02/app/oracle/gghome/19c/dirdat/lt
ADD RMTTRAIL /u02/app/oracle/gghome2/19c/dirdat/rt, EXTRACT shippump

```
GGSCI (ipfix01.example.com) 3> DBLOGIN USERIDALIAS gguser
Successfully logged into database.

GGSCI (ipfix01.example.com as ggadmin@DDSTST) 4> REGISTER EXTRACT shipep DATABASE

2025-08-25 15:50:23  INFO    OGG-02003  Extract SHIPEP successfully registered with database at SCN 3593212.


GGSCI (ipfix01.example.com as ggadmin@DDSTST) 5> ADD EXTRACT shipep, INTEGRATED TRANLOG, BEGIN NOW
EXTRACT (Integrated) added.


GGSCI (ipfix01.example.com as ggadmin@DDSTST) 6> ADD EXTTRAIL /u02/app/oracle/gghome/19c/dirdat/lt, EXTRACT shipep
EXTTRAIL added.

GGSCI (ipfix01.example.com as ggadmin@DDSTST) 7> ADD EXTRACT shippump, EXTTRAILSOURCE /u02/app/oracle/gghome/19c/dirdat/lt
EXTRACT added.


GGSCI (ipfix01.example.com as ggadmin@DDSTST) 8> ADD RMTTRAIL /u02/app/oracle/gghome2/19c/dirdat/rt, EXTRACT shippump
RMTTRAIL added.

GGSCI (ipfix01.example.com as ggadmin@DDSTST) 9>
```

DBLOGIN USERIDALIAS gguser2
ADD REPLICAT shipr, INTEGRATED, EXTTRAIL /u02/app/oracle/gghome2/19c/dirdat/rt

```
GGSCI (ipfix01.example.com) 2> DBLOGIN USERIDALIAS gguser2
Successfully logged into database.

GGSCI (ipfix01.example.com as ggadmin2@DDSTST) 3> edit param shipr



GGSCI (ipfix01.example.com as ggadmin2@DDSTST) 4> ADD REPLICAT shipr, INTEGRATED, EXTTRAIL /u02/app/oracle/gghome2/19c/dirdat/rt
REPLICAT (Integrated) added.


GGSCI (ipfix01.example.com as ggadmin2@DDSTST) 5>
```

And then start the Manager, extract, pump at source end as well as start Manager, Replicat at Target end.

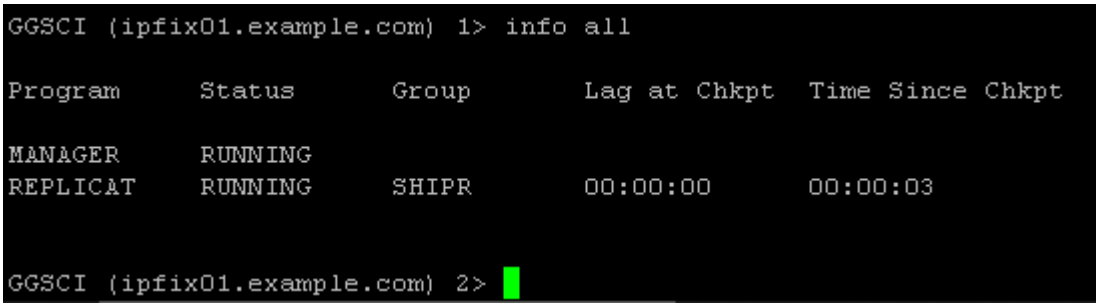
```
GGSCI (ipfix01.example.com) 6> info all

Program     Status      Group       Lag at Chkpt  Time Since Chkpt

MANAGER     RUNNING
EXTRACT     RUNNING     SHIPEP      00:00:00      00:00:07
EXTRACT     RUNNING     SHIPPUMP    00:00:00      00:00:06


GGSCI (ipfix01.example.com) 7>
```

Target end.


GGSCI (ipfix01.example.com) 1> info all
Program Status Group Lag at Chkpt Time Since Chkpt
MANAGER RUNNING
REPLICAT RUNNING SHIPR 00:00:00 00:00:03
GGSCI (ipfix01.example.com) 2>